

APPLIED PHYSICS FOR CSE STREAM(BPHYS102/202)

OFFICIAL LINKS

 Website

<https://braintube.site>

 WhatsApp Channel (VTU Notes & Updates)

<https://wa.me/918884624741>

 YouTube Channel

<https://youtube.com/@braintube-v9w?si>

MODULE-5

APPLICATIONS OF PHYSICS IN COMPUTING

PAPHYSICS OF ANIMATION

1. Physics of Animation – Introduction

Physics of Animation applies laws of motion to create **realistic movement** in computer graphics and games.

Physics ensures animations look **natural, believable, and consistent**.

Key areas:

- Motion and timing
- Forces and acceleration
- Size, scale, weight, and strength
- Graph-based motion analysis

2. Taxonomy of Physics-Based Animation Methods

Physics-based animation methods can be categorized as:

- **Kinematic methods:** Motion defined by position, velocity, acceleration
- **Dynamic methods:** Motion governed by forces and Newton's laws
- **Particle systems:** Used for fire, smoke, rain
- **Rigid body animation:** Solid objects with fixed shape
- **Soft body animation:** Deformable objects (cloth, skin)

3. Frames and Frames Per Second (FPS)

Frames

A **frame** is a single still image in an animation sequence.

Frames Per Second (FPS)

FPS is the number of frames displayed per second.

Common values:

- 24 FPS → Cinematic animation
- 30 FPS → Standard video
- 60 FPS → Smooth gaming animation

Higher FPS → smoother motion but higher computation cost.

4. Size and Scale

- **Size:** actual dimensions of objects
- **Scale:** relative size compared to surroundings

Correct scaling ensures:

- Objects appear realistic
- Motion matches visual mass

5. Weight and Strength

- **Weight** influences acceleration and impact
- Heavier objects accelerate slower
- Stronger characters lift or push heavier loads

Physics engines simulate weight using **mass** and **force**.

6. Motion and Timing in Animation

Motion quality depends on:

- Speed of movement
- Time taken to complete an action
- Acceleration and deceleration

Correct timing avoids:

- Jerky motion
- Unrealistic speed changes

7. Constant Force and Acceleration

Using Newton's second law:

$$F = ma$$

- Constant force → constant acceleration
- Used to simulate falling, jumping, pushing

Position update in animation:

$$x = ut + \frac{1}{2}at^2$$

8. The Odd Rule

Odd Rule states that:

- Odd numbers of frames look more **natural** than even numbers
- Helps avoid symmetry and artificial motion

Example:

- 5 frames → better than 4 frames for repeated actions

9. Odd-Rule Scenarios

- Walk cycles
- Repetitive hand motion
- Jump landing sequences

Odd timing prevents robotic repetition.

10. Motion Graphs

Motion graphs visually represent movement variables.

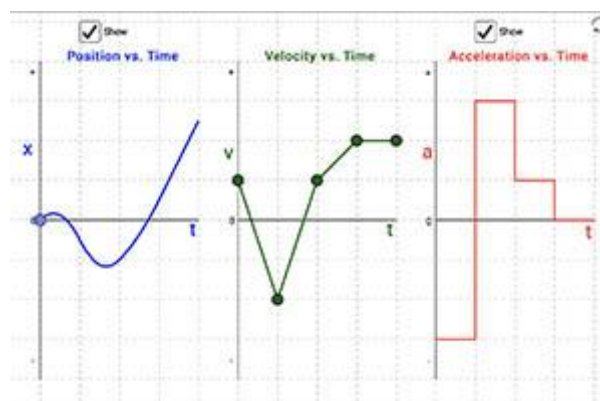
Types

- Position vs Time
- Velocity vs Time
- Acceleration vs Time

These graphs help animators:

- Debug motion
- Control smoothness
- Match physics laws

Motion Graphs



11. Character Animation – Jumping

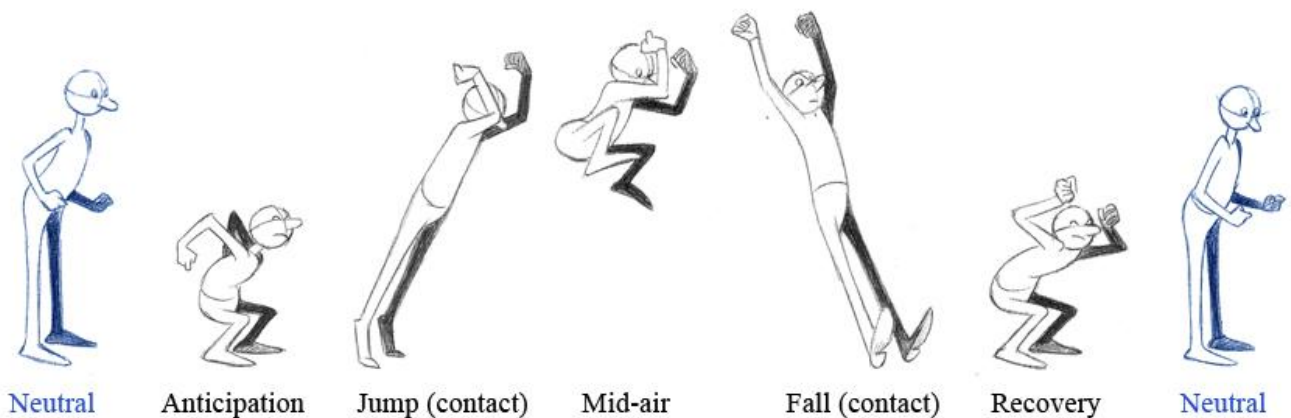
Jumping motion is divided into **distinct physical phases**.

Parts of a Jump

1. **Preparation (Crouch)**
2. **Take-off**
3. **Flight**
4. **Landing**

Each phase follows specific force and acceleration patterns.

Jump Phases Diagram (REQUIRED)



12. Jump Magnification

- Exaggerates jump height or hang time
- Makes motion visually appealing
- Often deviates slightly from real physics for effect

13. Stop Time

Stop time is a brief pause during impact (landing or take-off) to:

- Increase realism
- Convey weight
- Highlight force transfer

14. Walking Animation

Walking consists of:

- **Strides:** full cycle of movement
- **Steps:** individual foot placements

Correct stride length and timing avoid floating or sliding motion.

15. Walk Timing

Walk timing depends on:

- Character height
- Speed
- Ground friction

Incorrect timing leads to:

- Moon-walk effect
- Foot slipping

16. Numerical Problems (Typical)

Example

A character starts jumping upward with acceleration $a = 4 \text{ m/s}^2$ for $t = 1 \text{ s}$. Find displacement during take-off.

$$x = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(4)(1)^2 = 2 \text{ m}$$

STATISTICAL PHYSICS FOR COMPUTING

1. Introduction to Statistical Physics

Statistical physics applies probability and statistics to analyze systems containing a large number of particles or events.

In computing, it is used for:

- Data analysis
- Performance modeling
- Randomized algorithms
- Simulation and uncertainty estimation

2. Descriptive and Inferential Statistics

2.1 Descriptive Statistics

Used to **summarize and describe** data.

Common measures:

- **Mean:** $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$
- **Median**
- **Mode**

- **Variance:** $\sigma^2 = \frac{1}{N} \sum (x_i - \bar{x})^2$
- **Standard deviation:** $\sigma = \sqrt{\sigma^2}$

2.2 Inferential Statistics

Used to **draw conclusions** about a population from a sample.

Includes:

- Estimation
- Hypothesis testing
- Confidence intervals
- Predictive modeling

3. Poisson Distribution

3.1 Definition

The **Poisson distribution** gives the probability of observing exactly x events in a fixed interval, when events occur **independently** with a known average rate.

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

where

λ = mean number of occurrences.

3.2 Properties

- Mean = Variance = λ
- Suitable for **rare events**
- Discrete probability distribution

4. Modeling the Probability of Proton Decay

Proton decay is an extremely **rare event**.

Poisson distribution is suitable because:

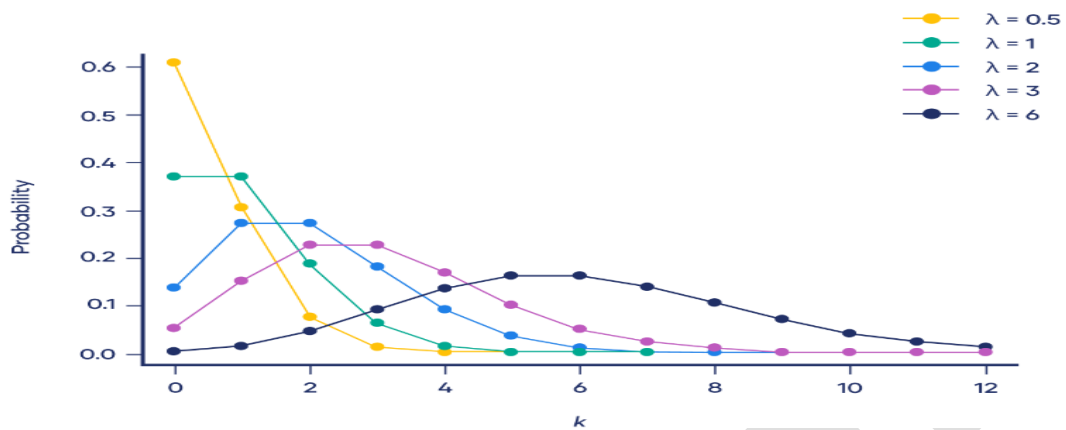
- Events are random and independent
- Average rate is very small
- Probability over long observation time is required

If λ is the expected number of decays:

$$P(0) = e^{-\lambda}$$

gives the probability that **no decay occurs** in a given time.

Poisson Distribution Curve



5. Normal Distribution (Gaussian Distribution)

5.1 Definition

The **normal distribution** is a continuous probability distribution defined by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(x-\mu)^2}{2\sigma^2} \right]$$

where

μ = mean,

σ = standard deviation.

5.2 Properties

- Symmetric bell-shaped curve
- Mean = Median = Mode
- Total area under curve = 1
- Widely appears due to **central limit theorem**

6. Applications of Normal Distribution in Computing

- Measurement noise modeling
- Error analysis
- Performance metrics
- Machine learning assumptions

7. Monte Carlo Method

7.1 Principle

The **Monte Carlo method** uses **random sampling** to solve numerical problems that are difficult to compute analytically.

Key idea:

- Random points are generated
- Statistical averaging is used to estimate results

8. Determination of the Value of π (Monte Carlo Method)

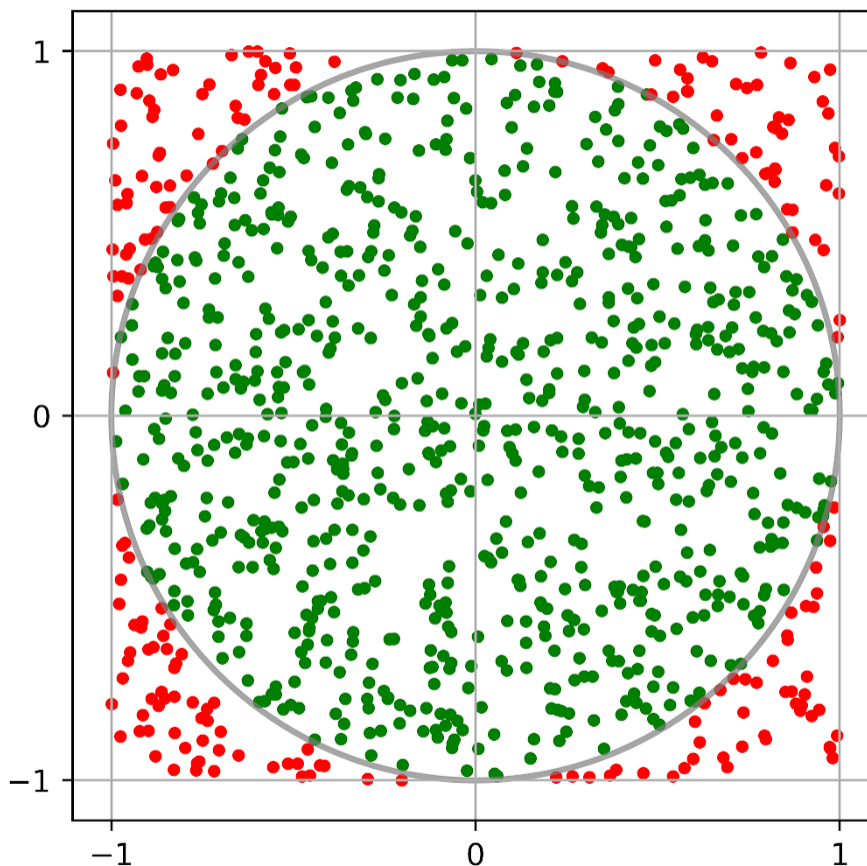
Method

- Consider a square of side 2 units
- Inscribe a circle of radius 1 unit
- Generate random points within the square
- Count points inside the circle

$$\pi \approx 4 \times \frac{\text{Points inside circle}}{\text{Total points}}$$

Monte Carlo Estimation of π

1000 Samples
 $\pi \approx 3.164$



9. Numerical Problems (Typical)

Example 1 (Poisson)

If the mean number of events is $\lambda = 2$, find $P(0)$.

$$P(0) = e^{-2} = 0.1353$$

Example 2 (Monte Carlo – Conceptual)

If 785 points fall inside a circle out of 1000 random points:

$$\pi \approx 4 \times \frac{785}{1000} = 3.14$$